

DevOps & Cloud Deployment – Module 2 Exercise

Hands-on Exercise No. 2

Course: DevOps & Cloud Deployment

Module: CI/CD Pipelines

Total Marks: 10

Due Date: 15/06/2026

Instructions

Prepare your solution in MS Word.

Submit in .doc or .docx format.

Late or copied submissions will not be accepted.

Learning Outcome

After completing this exercise, you will be able to:

- Understand Continuous Integration and Continuous Deployment concepts.
 - Configure a CI/CD pipeline using Jenkins or GitHub Actions.
 - Automate build, test, and deployment processes.
 - Monitor pipeline execution and troubleshoot errors.
-

Problem Statement

You are working as a DevOps Engineer for a software company. The client wants an automated pipeline that builds and deploys a Node.js application whenever new code is pushed to GitHub.

Task Requirements

Create a simple Node.js application (`app.js`) with a basic route.

Push the project to a GitHub repository.

Configure a CI/CD pipeline using either:

- GitHub Actions (`.github/workflows/ci.yml`)
OR

- Jenkins Pipeline script

The pipeline should include the following steps:

- Install dependencies using `npm install`
- Run tests using `npm test`
- Deploy the application (can be simulated using a console message or deployment script)

Trigger the pipeline by pushing code changes to the repository.

Take Screenshots of

- CI/CD pipeline configuration file.
 - Successful pipeline execution.
 - Deployment output or logs.
-

Submission Guidelines

Upload your .docx file with pipeline configuration, screenshots, GitHub link, and explanations via LMS.